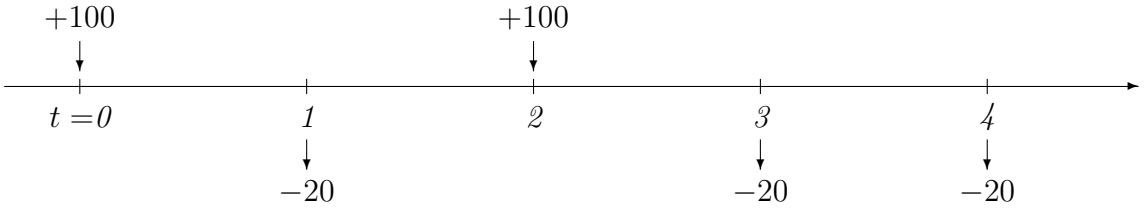
F78AA - Actuarial and Financial Mathematics A

Tutorial 2 – Valuing cashflows and annuities

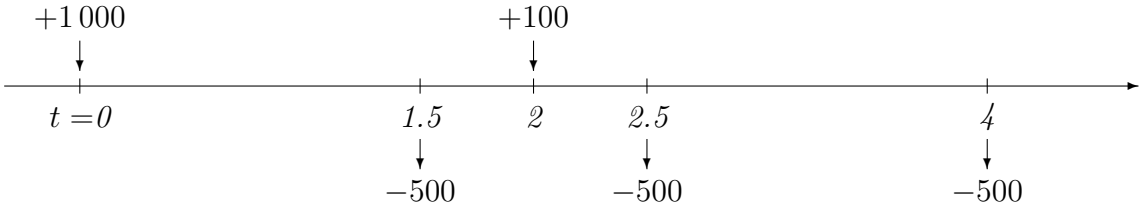
Question 1 The effective rate of interest is 6% per time unit. Cashflows are as follows (the time axis is calibrated in time units):



For these cashflows, calculate:

1. the accumulation at time $t = 5$ time units.
2. the accumulation at time $t = 10$ time units.
3. the present value at time $t = 0$ time units.

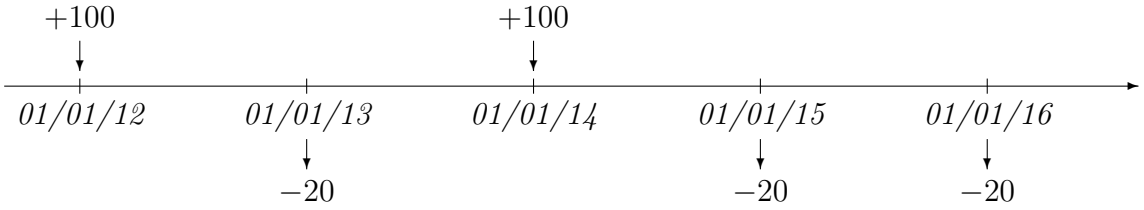
Question 2 The effective rate of interest is 8% per annum. Cashflows are as follows (the time axis is calibrated in years):



For these cashflows, calculate:

1. the total accumulation at time $t = 5$ years.
2. the total present value at time $t = 0$ years.
3. the total value at time $t = 2$ years.

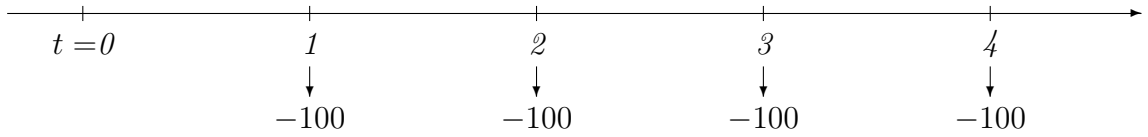
Question 3 The effective rate of interest is 3% per annum during 2012 and 2013, 2% per half-year until 1 July 2015, and 1% per month thereafter. Cashflows are as follows (the time axis is calibrated in calendar time):



For these cashflows, calculate the total:

1. present value on 01/01/12.
2. accumulation on 01/05/17.
3. value on 01/07/15.

Question 4 The effective rate of interest is 5.5% per time unit. You have a liability to pay the cashflows as shown in the following diagram (the time axis is calibrated in time units) at the end of the 1st, 2nd, 3rd and 4th time units:



In order to discharge this liability, you set up a bank account time $t = 0$ and pay into it an amount £X. You will meet the future payments out of this bank account.

Calculate the value of X that will allow you to exactly meet all the future payments.

Question 5

1. Describe in words the meaning of the following symbols:

   (a) $s^{i}_{\overline{10|}}$ (b) $\ddot{s}^{i}_{\overline{25|}}$ (c) $a^{i}_{\overline{10|}}$ (d) $\ddot{a}^{i}_{\overline{20|}}$

2. Assuming $i = 0.07$ per time unit, calculate the values of the above symbols.

Question 6 Prove the following:

1. $s_{\overline{n|}} = (1+i)^n a_{\overline{n|}}$.
2. $s_{\overline{n|}} = 1 + \ddot{s}_{\overline{n-1|}}$.
3. $\ddot{a}_{\overline{n|}} = v^n \ddot{s}_{\overline{n|}}$.
4. $\ddot{a}_{\overline{n|}} = 1 + a_{\overline{n-1|}}$.

Solutions: Q1.1. 184.00; Q1.2. 246.24; Q1.3. 137.50; Q2.1 -205.35; Q2.2 -139.75; Q2.3 -163.01; Q3.1 139.99; Q3.2 196.17; Q3.3 157.60; Q4.1 350.51; Q5.2 (a) 13.81645; Q5.2 (b) 67.67647; Q5.2 (c) 7.02358; Q5.2 (d) 11.33560.